Exp. 4. Band Structure Simulation for Graphene and CNTs using NanoHub

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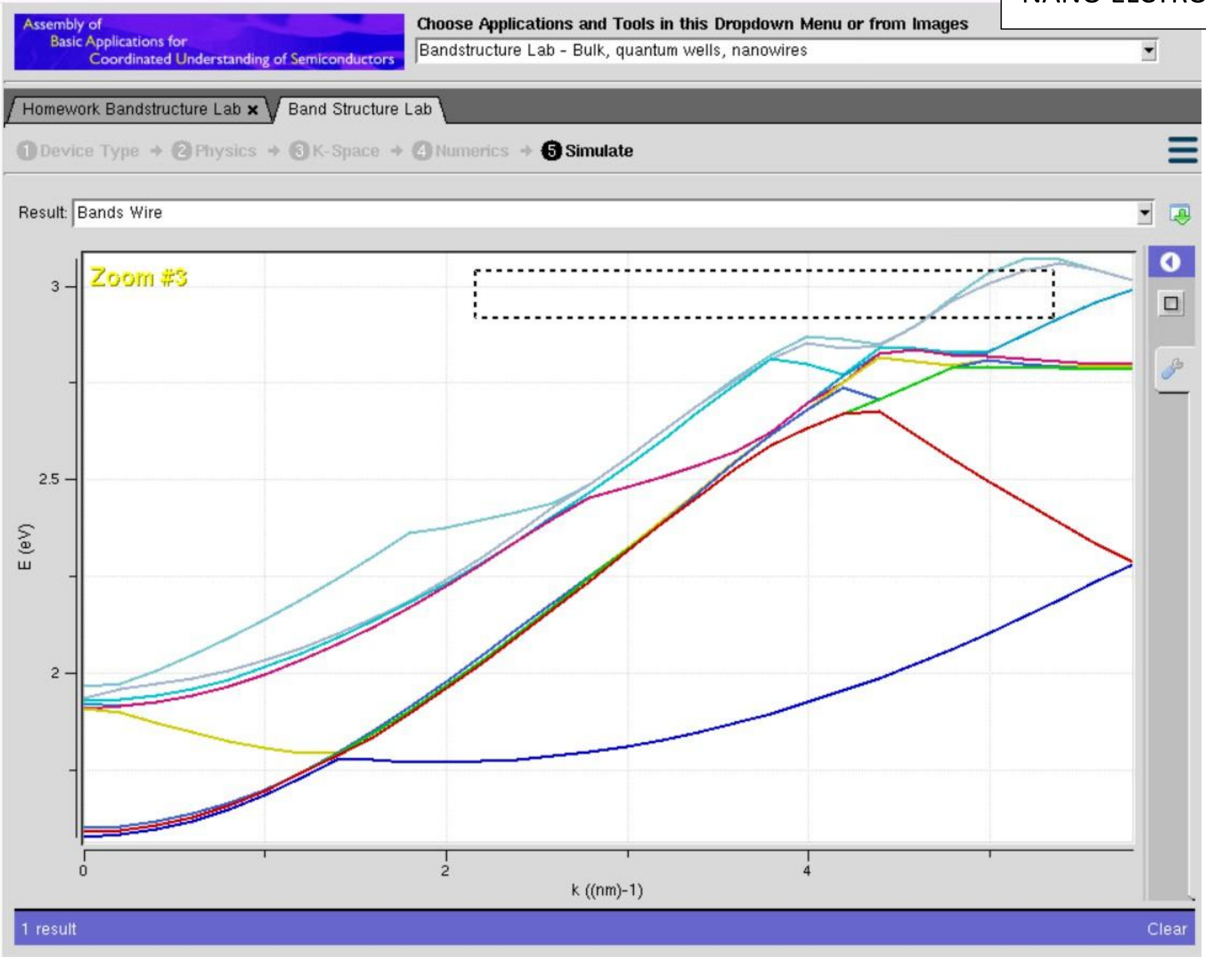
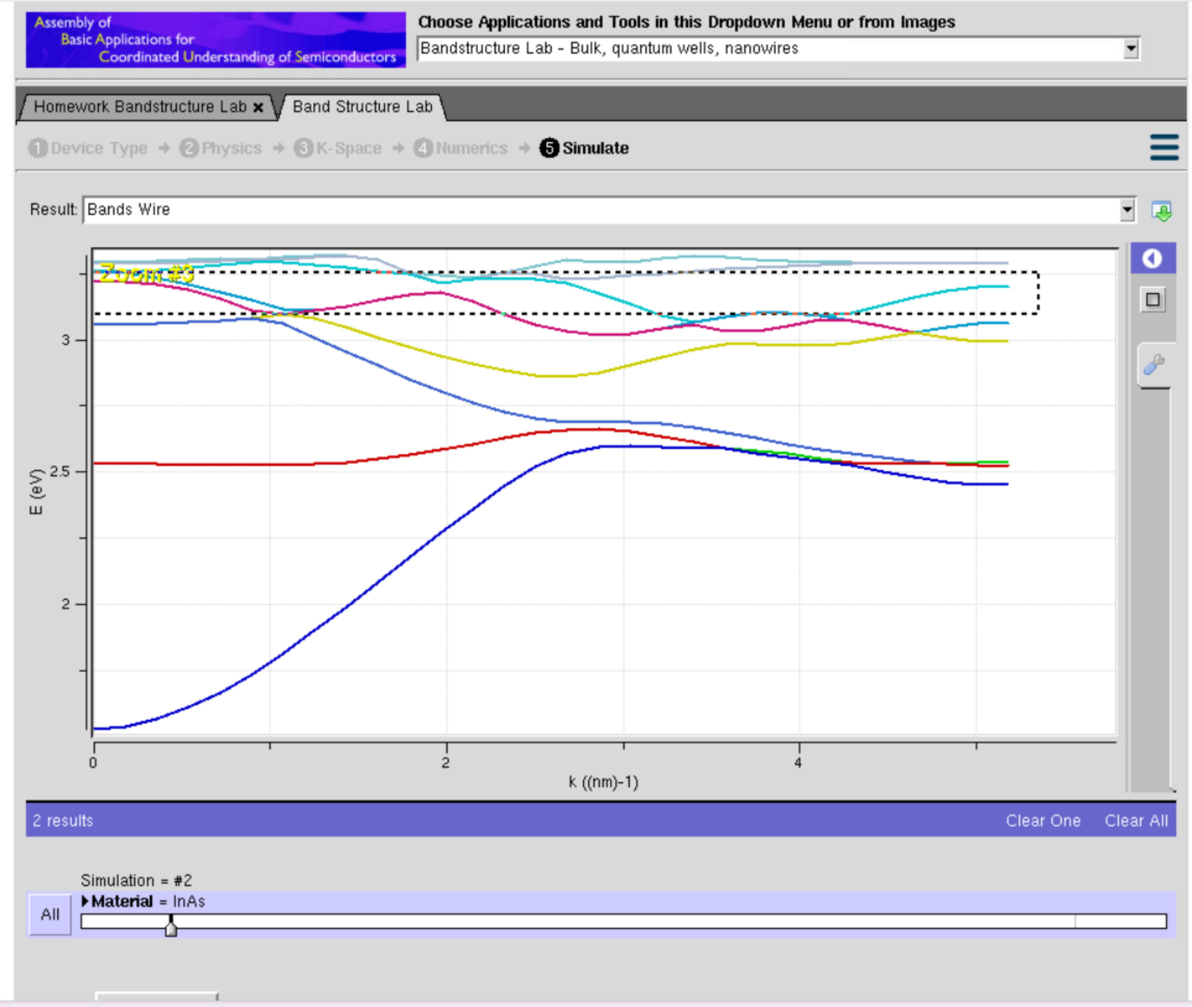


This graph illustrates the relationship between Energy (E, eV**)** and Wavevector (k, nm^{-1}) for a 1D semiconductor nanowire simulated using the ABACUS tool in nanoHUB. Due to quantum confinement in two spatial directions, the continuous energy bands of the bulk material split into discrete one-dimensional subbands.

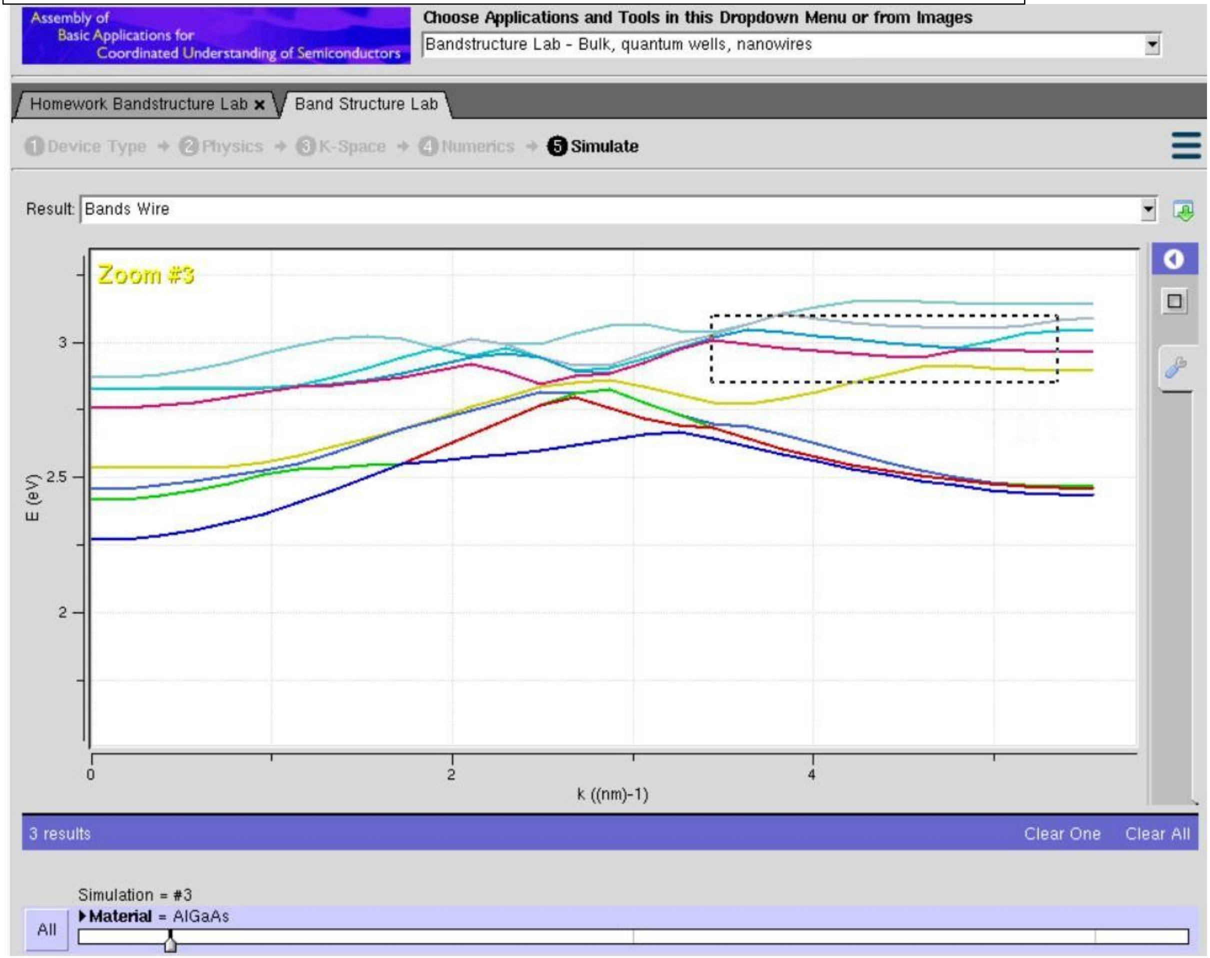
The curvature of the energy bands indicates the effective mass of charge carriers. Steeper parabolic bands correspond to a lower effective mass, allowing electrons to move more rapidly, whereas flatter bands, particularly near the upper-right region above 2.8 eV, indicate a higher effective mass and an increased density of states. The subband minima at k = 0 (approximately 1.8 eV**)** define the onset of conduction, while the band intersections observed around k = 3–4 nm^{-1} suggest possible inter-subband scattering, which plays a significant role in understanding electron transport in nanoscale semiconductor devices.

Case Study 1: Simulate the band-structure of InAs circular nanowire



Case Study 1 examines the band structure of an Indium Arsenide (InAs) circular nanowire using the ABACUS Band Structure Lab available on nanoHUB**.** The objective is to investigate the influence of the material's low electron effective mass under one-dimensional quantum confinement. This study is important for the development of high-electron-mobility nanoscale transistors, where quantum confinement strongly affects carrier transport. The simulation is performed by defining a 1D circular nanowire geometry, selecting InAs as the semiconductor material, and computing the energy–wavevector (E–k) dispersion relation. The resulting band structure exhibits widely separated, strongly curved energy subbands, reflecting the light effective mass and enhanced electron mobility characteristic of InAs.

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



Case Study 2 investigates the band structure of an Aluminum Gallium Arsenide (AlGaAs) circular nanowire using the ABACUS tool on nanoHUB. The study focuses on how variations in the molar fraction and the material's relatively higher electron effective mass influence the bandgap and the distribution of energy subbands. This analysis is important for the design of heterostructure nanowires and quantum-based optoelectronic devices**.** The simulation is carried out by selecting a 1D circular nanowire geometry, choosing AlGaAs as the semiconductor material, assigning the required molar fraction, and calculating the electronic band structure**.** The results show closely spaced and densely packed energy subbands**,** which are characteristic of the higher carrier effective mass in AlGaAs.

In conclusion**,** reducing the dimensions of a semiconductor to form a one-dimensional (1D) nanowire results in strong quantum confinement, which transforms continuous bulk energy bands into discrete quantized subbands. By simulating the Energy–Wavevector (E–k) band **s**tructure using nanoHUB, researchers can better understand and tailor these subband characteristics. This insight is essential for improving carrier transport and enhancing the performance of future nanoscale electronic and optoelectronic devices.